AquaSentry Autonomous Catamaran Prototype Blueprint

Open-Hardware Water Surveillance Vessel & Arsenic Stripping Cell

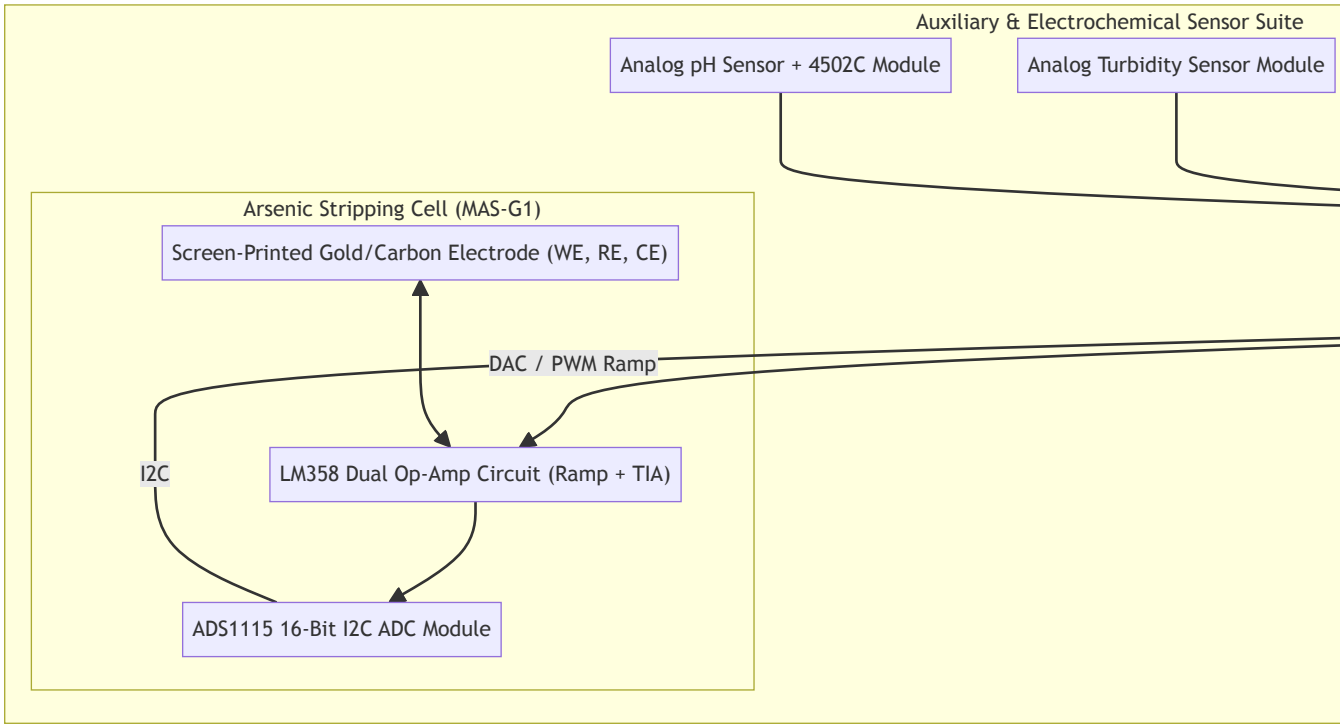


Executive Architecture Overview

The **AquaSentry Autonomous Surface Vessel (ASV)** is a twin-hull catamaran robot engineered to screen lakes, rivers, and community reservoirs for dissolved arsenic (As^{3+}/As^{5+}) and geochemical interference parameters (pH, TDS/EC, Turbidity, Temperature).

To ensure anyone can construct, test, and deploy this vessel without proprietary industrial tooling, **over 90% of the platform utilizes standard, off-the-shelf robotics hobbyist modules**, standard **2020 aluminum extrusions**, and a **desktop 3D printer** for the hull structure.

System Architecture Diagram



Component-by-Component Bill of Materials (BOM)

1. Hull & Mechanical Structure

Part Name	Role in Robot	Why Chosen / Standard Availability
Twin 3D-Printed Pontoons	Provides buoyancy, hydrodynamic low-drag displacement, and rollover stability during lake chop.	Printable on standard 220×220×250 mm 3D printers in 3 interlocking modular segments (Bow, Mid, Stern).
2020 Aluminum Extrusions (2x 450mm)	Heavy-duty crossbeams connecting the two pontoons into a rigid catamaran chassis.	Standard 3D printer extrusion; slotted for infinitely adjustable mounting brackets using M4 T-nuts.
IP67 Waterproof Junction Box (150×200×75mm with clear lid)	Houses the ESP32, motor driver, ADC, and wiring, protecting electronics from lake spray and rain.	Off-the-shelf electrical junction box costing under $6; clear lid allows visual inspection of status LEDs.
Ballast Fin & Weight (M8/M10 bolts or lead fishing weights)	Mounted to the keel bottom below the center of mass.	Lowers center of gravity to prevent capsize in high winds; self-righting physics.

2. Propulsion & Steering (Zero-Leak Differential Drive)

Part Name	Role in Robot	Why Chosen / Standard Availability
2x 12V 775 DC Motors + Waterproof Shaft Kits *(or 2x T60 Underwater Thrusters)*	Submerged stern propellers providing forward, reverse, and differential steering.	Ubiquitous in RC boats and DIY robotics. Avoiding traditional mechanical rudders eliminates rudder shaft seal leaks.
L298N Dual H-Bridge Motor Driver *(or IBT-2 / BTS7960)*	Drives and modulates speed of both thrusters via PWM from the microcontroller.	Most common motor driver in maker robotics; handles up to 2A continuous per channel with heat sink.

3. Fluidic Sampling Subsystem (Non-Immersion Cell)

Part Name	Role in Robot	Why Chosen / Standard Availability
12V Micro Peristaltic Dosing Pump	Draws a precise 15 mL water sample through silicone tubing into the internal analysis chamber, then reverses to purge.	Standard medical/aquarium dosing pump (7–10). Self-priming, positive displacement; liquid only touches the tubing (no contamination).
3D-Printed Reaction Well (with silicone O-ring)	Hermetically sealed 15 mL reaction chamber holding the sample and electrode strip.	Keeps stripping electrodes completely isolated from open-water waves, debris, and biofouling.
Silicone Tubing (2mm ID / 4mm OD, 1 meter)	Fluid transport from lake intake filter to reaction well and exhaust.	Highly flexible, chemical-resistant, zero leaching.

4. Sensor Suite & Electrochemical Core

Part Name	Role in Robot	Why Chosen / Standard Availability
Screen-Printed 3-Electrode Strip *(MAS-G1 / Au-Carbon)*	**[Specialized Component]** Gold working electrode (Au-WE), Ag/AgCl reference (RE), carbon counter (CE) for Anodic Stripping Voltammetry (ASV).	**The only non-standard sensor part.** Screen-printed disposable strips (Metrohm DropSens or HKU MAS-G1) provide repeatable micro-gap voltammetry.
LM358 / MCP6002 Op-Amp Circuit	Forms the micro-potentiostat: applies deposition potential (-0.8V) and converts stripping nanoamp currents to voltages via Transimpedance Amplification (TIA).	Standard $0.20 operational amplifier. Simple circuit schematic easily assembled on perfboard.
ADS1115 16-Bit I2C ADC Module	Reads the analog output voltage from the potentiostat circuit with high precision (down to $7.8\ \mu\text{V}$ per LSB).	Built-in Arduino/ESP32 ADCs (10-12 bit with noise) cannot resolve nanoamp arsenic stripping peaks. ADS1115 is standard in maker electronics.
DS18B20 Waterproof Temperature Probe	Measures surface water temperature in real-time.	Stainless steel probe capsule; temperature dictates electrochemical stripping kinetic rates and feeds the neural classifier.
Analog TDS / Electrical Conductivity Sensor	Measures dissolved ionic minerals ($\mu\text{S}/\text{cm}$).	Standard 5V DFRobot-style conductivity probe; acts as proxy for background ionic interference.
Analog Turbidity Sensor Module	Measures water clarity / suspended silt in NTU via optical infrared scattering.	Suspended clay particles adsorb arsenic ions; required by AquaNet-V1 neural model.
Analog pH Sensor (E-201-C Probe + 4502C Module)	Quantifies water acidity ($pH\ 5.5 - 9.5$).	Arsenic speciation (As^{3+} arsenite vs As^{5+} arsenate) is strictly pH-dependent.

5. Brain, Navigation & Telemetry

Part Name	Role in Robot	Why Chosen / Standard Availability
ESP32 DevKit V1 (30-Pin)	Primary robot microcontroller running sensor polling, pump sequencing, neural edge inference, and communications.	Dual 240MHz cores, built-in Bluetooth and Wi-Fi, 3.3V logic matching LoRa/GPS, cost < $5.
NEO-6M / NEO-8M GPS Module	Synchronizes real-time latitude/longitude coordinates to every water reading for spatial mapping.	Standard UART GPS with active ceramic antenna; lock acquisition in under 35 seconds.
SX1278 LoRa Transceiver (433MHz or 915MHz)	Transmits compressed telemetry packets to a shore station over 2–5 km range.	Zero recurring cellular SIM costs; ideal for remote rural lakes and reservoirs.

6. Power Distribution

Part Name	Role in Robot	Why Chosen / Standard Availability
3S 18650 Li-Ion Battery Pack (11.1V, 4400–5200 mAh with BMS)	Provides system power for up to 8–12 hours of active patrol and sampling.	Standard high-density cells; fits cleanly inside the central hull cavity.
LM2596 DC-DC Step-Down Buck Converter	Steps 11.1V down to regulated 5.0V for logic boards, sensors, and GPS.	High efficiency (up to 92%), adjustable via onboard potentiometer.
5W 6V Monocrystalline Solar Panel + TP4056/CN3791 *(Optional)*	Mounted on deck; trickle charges the battery during sunlight hours.	Extends autonomous mission duration indefinitely in calm tropical conditions.

3D Printing Specifications for the Shell

To achieve watertight integrity and buoyancy on a standard hobbyist 3D printer (e.g. Ender 3, Bambu Lab, Prusa MK4):

1. **Filament Selection**:
2. **PETG** (*Recommended*): Excellent layer adhesion, high UV resistance, and completely waterproof when printed with slightly higher temperatures ($240^{\circ}\mathrm{C}$).
3. **PLA+**: Acceptable if coated externally with a two-part marine epoxy (e.g., XTC-3D or clear marine polyurethane varnish).
4. **Slicer Settings for Waterproofing**:
5. **Perimeters / Wall Line Count**: **4 to 5 walls** (prevents water weeping through microscopic layer seams).
6. **Top/Bottom Solid Layers**: **5 layers minimum**.
7. **Infill Density & Pattern**: **25%–30% Gyroid**. Gyroid infill provides omnidirectional strength and creates buoyant air pockets even if an outer wall sustains damage.
8. **Layer Height**: **0.20 mm** (smoother interlayer bond than 0.28 mm).
9. **Modular Hull Assembly**:
10. Each pontoon measures ~600–700 mm total length, split into 3 segments:
    1. **Bow Section**: Hydrodynamic raked bulbous bow to slice through surface weeds.
    2. **Mid Section**: Contains M4 captive nut slots for mounting the 2020 aluminum crossbars.
    3. **Stern Section**: Recessed bracket to clamp the thruster motor and guide wires.
11. Join the 3 sections using stainless steel M4 threaded rods passing through internal alignment channels, sealed with silicone gasket sealant between mating faces.

Operating Mission Routine (Autonomous Screening Cycle)

1. **Autonomous Waypoint Patrol**:
2. GPS guides the catamaran along predefined survey coordinates using differential thrust.
3. **Station-Keeping & Sample Intake**:
4. Vessel stops at target waypoint.
5. Peristaltic pump runs forward for 8 seconds, drawing 15 mL of raw water into the sealed reaction well.
6. **Multi-Parameter Pre-Scan**:
7. Submerged keel probes record pH, conductivity (EC), turbidity, and temperature.
8. **Anodic Stripping Voltammetry (ASV) Quantification**:
9. Potentiostat applies a deposition potential ($-0.8\ \mathrm{V}$) for 60 seconds to pre-concentrate As^{3+} onto the gold working electrode.
10. Potential ramps anodically toward $+0.4\ \mathrm{V}$ while measuring stripping oxidation current (i_p) at $\sim +0.15\ \mathrm{V}$ via the ADS1115.
11. Savitzky-Golay quadratic filtering and baseline subtraction calculate peak net current ($\mu\mathrm{A}$) to determine arsenic concentration ($\mathrm{mg/L}$).
12. **AquaNet-V1 Edge Inference**:
13. ESP32 runs the 5-8-6-4 feed-forward neural forward pass locally in $< 2\ \mathrm{ms}$, categorizing risk into Safe, Caution, Unsafe, or Hazardous.
14. **Telemetry Transmission & Purge**:
15. Telemetry string formatted: `AS:0.042,PH:7.2,T:24.5,EC:380,TURB:4.2,LAT:25.61,LON:85.14`
16. Sent via LoRa / Bluetooth to shore receiver and automatically ingested by `bridge/serial_bridge.py`.
17. Peristaltic pump reverses for 8 seconds, purging the reaction well into the stern wake.
18. Catamaran proceeds to next GPS waypoint.

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